

YOUSEF ELSHAER

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EDUCATION

University of Guelph

Bachelor of Computing (Co-op), Major in Computer Science

Guelph, Ontario

Sep 2025 – Apr 2030

SKILLS

Programming Languages: C, Python, C++, HTML, CSS, JavaScript, SQL

Tools & Technologies: Git, GitHub, Linux, VS Code, JetBrains, pytest, REST APIs

Frameworks & Libraries: FastAPI, SQLAlchemy, OpenCV, YOLO

Soft Skills: Customer Service, Teamwork, Leadership, Time Management, Adaptability

PROJECTS

Sentinel | Python, FastAPI, SQLAlchemy, REST APIs, SQL [GitHub](#) | [Live demo](#)

Aug 2026

- Developed a Python-based cybersecurity platform to analyze IP addresses, domains, URLs, file hashes, and CVEs using multiple threat-intelligence APIs.
- Integrated VirusTotal, AbuseIPDB, URLhaus, and CISA vulnerability data through REST APIs and normalized responses into a unified analysis system.
- Designed a weighted risk-scoring engine that correlates threat indicators and classifies results as Low, Medium, High, or Critical.
- Built REST API endpoints with FastAPI and implemented persistent investigation history using SQLAlchemy and a relational database.
- Implemented input validation, API error handling, caching, rate-limit handling, and automated testing with pytest.

Computer Vision Security Monitoring System | Python, OpenCV, YOLO

Present

- Developing a real-time security monitoring system using **OpenCV and YOLO** for person, motion, and object detection.
- Implementing **custom detection zones, time-based rules, and threat scoring** to identify suspicious activity.
- Building automated **alerts and CSV/JSON event logging** with API integration for security notifications.
- Designing a **multi-threaded processing pipeline** to handle video capture, object detection, threat analysis, and alert generation efficiently in real time.

Personal Portfolio Website | HTML, CSS, JavaScript [GitHub](#)

Sep 2026

- Developed and deployed a responsive personal portfolio website to showcase software projects, technical skills, and professional experience.
- Built interactive project sections with direct links to GitHub repositories and professional profiles.
- Designed a responsive interface optimized for desktop and mobile devices and deployed it on a custom domain.

EXPERIENCE

Summer Camp Counselor

YMCA

Jun 2025 – Present

Mississauga, Ontario

- Plan and lead structured programs for children in sports, arts, and science
- Supervise daily activities to maintain a safe, positive, and inclusive environment
- Collaborate with fellow counselors to design and deliver engaging, age-appropriate programming

Sales Agent

InnovaTech Solutions

Sep 2024 – Oct 2025

Mississauga, Ontario

- Assisted in selling SEO services by explaining basic online marketing benefits to clients
- Supported the sales team with outreach, follow-ups, and tracking client interest